NATIONAL ECONOMICS UNIVERSITY
COLLEGE OF TECHNOLOGY

REPORT

SUBJECT: STATISTICAL LEARNING

Topic:

PREDICTING CUSTOMER INSURANCE PURCHASE BEHAVIOR

Group 5: HÀ THU HÀ - 11248476
Class: DS66A
Supervisor: Dr. Thien Van Luong

Hanoi, October 30, 2026

ĐÁNH GIÁ QUYỂN ĐỒ ÁN TỐT NGHIỆP

(Dùng cho giảng viên hướng dẫn)

Tên giảng viên đánh giá: .....

Họ và tên sinh viên: ..... MSSV: .....

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Chọn các mức điểm phù hợp cho sinh viên trình bày theo các tiêu chí dưới đây:

Rất kém (1); Kém(2); Đạt(3); Giỏi(4); Xuất sắc(5)

Có sự kết hợp giữa lý thuyết và thực hành (20)						
1	Nêu rõ tính cấp thiết và quan trọng của đề tài, các vấn đề và các giả thuyết (bao gồm mục đích và tính phù hợp) cũng như phạm vi ứng dụng của đồ án	1	2	3	4	5
2	Cập nhật kết quả nghiên cứu gần đây nhất (trong nước/quốc tế)	1	2	3	4	5
3	Nêu rõ và chi tiết phương pháp nghiên cứu/giải quyết vấn đề	1	2	3	4	5
4	Có kết quả mô phỏng/thực nghiệm và trình bày rõ ràng kết quả đạt được	1	2	3	4	5
Có khả năng phân tích và đánh giá kết quả (15)						
5	Kế hoạch làm việc rõ ràng bao gồm mục tiêu và phương pháp thực hiện dựa trên kết quả nghiên cứu lý thuyết một cách có hệ thống	1	2	3	4	5
6	Kết quả được trình bày một cách logic và dễ hiểu, tất cả kết quả đều được phân tích và đánh giá thỏa đáng	1	2	3	4	5
7	Trong phần kết luận, tác giả chỉ rõ sự khác biệt (nếu có) giữa kết quả đạt được và mục tiêu ban đầu đề ra đồng thời cung cấp lập luận để đề xuất hướng giải quyết có thể thực hiện trong tương lai	1	2	3	4	5
Kỹ năng viết quyển đồ án (10)						
8	Đồ án trình bày đúng mẫu quy định với cấu trúc các chương logic và đẹp mắt (bảng biểu, hình ảnh rõ ràng, có tiêu đề, được đánh số thứ tự và được giải thích hay đề cập đến; căn lề thống nhất, có dấu cách sau dấu chấm, dấu phẩy v.v.), có mở đầu chương và kết luận chương, có liệt kê tài liệu tham khảo và có trích dẫn đúng quy định	1	2	3	4	5
9	Kỹ năng viết xuất sắc (cấu trúc câu chuẩn, văn phong khoa học, lập luận logic và có cơ sở, từ vựng sử dụng phù hợp v.v.)	1	2	3	4	5
Thành tựu nghiên cứu khoa học (5) *(chọn 1 trong 3 trường hợp)*						
10a	Có bài báo khoa học được đăng hoặc chấp nhận đăng/Đạt giải SVNCKH giải 3 cấp Viện trở lên/Có giải thưởng khoa học (quốc tế hoặc trong nước) từ giải 3 trở lên/Có đăng ký bằng phát minh, sáng chế	5				
10b	Được báo cáo tại hội đồng cấp Viện trong hội nghị SVNCKH nhưng không đạt giải từ giải 3 trở lên/Đạt giải khuyến khích trong các kỳ thi quốc gia và quốc tế khác về chuyên ngành (VD: TI contest)	2				
10c	Không có thành tích về nghiên cứu khoa học	0				
Điểm tổng		/50				
Điểm tổng quy đổi về thang 10						

Nhận xét khác *(về thái độ và tinh thần làm việc của sinh viên)*

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Ngày: ... / ... / 20...

Người nhận xét

(Ký và ghi rõ họ tên)

ĐÁNH GIÁ QUYỂN ĐỒ ÁN TỐT NGHIỆP

(Dùng cho cán bộ phản biện)

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Điểm tổng quy đổi về thang 10						

Nhận xét khác của cán bộ phản biện

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Người nhận xét

(Ký và ghi rõ họ tên)

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation / Symbol	Full name	Meaning in this report
– General and dataset		
ML	Machine Learning	Learning of predictive models from data
CoIL	Computational Intelligence and Learning	Challenge 2000 that provided the benchmark dataset
COIL 2000	Insurance Company Benchmark (COIL 2000)	Dataset of 5,822 training and 4,000 test customers
UCI	University of California, Irvine	Repository (KDD archive) from which the data are obtained
EDA	Exploratory Data Analysis	Analysis of data quality and predictor–target relationships
`CARAVAN`	Caravan policy indicator	Binary target variable (1 = owns a caravan policy)
`M`, `P`, `A`	Variable groups	Socio-demographic, premium and policy-count predictors, respectively
– Models		
LR	Logistic Regression	Linear probabilistic classifier
KNN	K-Nearest Neighbours	Distance-based, instance-based classifier
DT	Decision Tree	Rule-based classifier using sequential splits
RF	Random Forest	Ensemble of bootstrap-trained decision trees
SVM	Support Vector Machine	Maximum-margin classifier
RBF	Radial Basis Function	Nonlinear SVM kernel
L1, L2	L1 (Lasso) and L2 (Ridge) penalties	Regularization terms on coefficient magnitudes
– Feature analysis and preprocessing		
BH	Benjamini–Hochberg procedure	Multiple-testing adjustment of p-values

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Abbreviation / Symbol	Full name	Meaning in this report
FDR	False Discovery Rate	Expected proportion of false discoveries among significant results
IV	Information Value	Measure of the separation between owners and non-owners provided by a predictor
WOE	Weight of Evidence	Log-ratio of the class-conditional level proportions, used to compute IV
OHE	One-Hot Encoding	Representation of a categorical level by a binary indicator column
E1, E2, E3	Feature-selection experiments	High-IV, high+medium-IV and all significant predictors, respectively
– Evaluation and statistics		
TP	True Positive	Owner correctly predicted as owner
FP	False Positive	Non-owner incorrectly predicted as owner
TN	True Negative	Non-owner correctly predicted as non-owner
FN	False Negative	Owner incorrectly predicted as non-owner
TPR	True Positive Rate	$TP/(TP+FN)$, also called recall
FPR	False Positive Rate	$FP/(FP+TN)$
AP	Average Precision	Primary metric; summary of the precision–recall curve
ROC-AUC	Area Under the Receiver Operating Characteristic Curve	Secondary, threshold-independent ranking metric
F1	F1-score	Harmonic mean of precision and recall
– Mathematical notation		
$\mathscr{D}$	Labelled dataset	$\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$
N, n	Number of observations	Size of the dataset or training set
$\mathbf{x}_i$	Input feature vector	85 attributes of customer i
y_i	Target variable	Caravan policy ownership of customer i (0 or 1)
$f(\cdot)$, s_i	Scoring function and score	Predicted score or probability of ownership of customer i

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Abbreviation / Symbol	**Full name**	**Meaning in this report**
$P(\cdot)$, $\hat{P}(\cdot)$	Probability and estimated probability	Population and estimated (observed) proportions
H_0, H_1	Null and alternative hypotheses	Independence or dependence of a predictor and the target
α	Significance level	Threshold for the p-values (0.05)
q_j	Adjusted q-value	BH-adjusted p-value of predictor j
r_j	Spearman correlation	Correlation between predictor levels and their ownership rates
β_0, β_j	Regression coefficients	Intercept and coefficients of Logistic Regression
$\sigma(\cdot)$	Sigmoid function	$1/(1+\exp(-z))$
λ, C	Regularization strength and its inverse	$C \propto 1/\lambda$; smaller C means stronger regularization
k	Number of neighbours	Neighbours used by KNN
d_E, d_M	Euclidean and Manhattan distances	Distance measures used by KNN
$I(t)$, $\text{Gini}(t)$, $H(t)$	Node impurity, Gini impurity, entropy	Split criteria of the Decision Tree
B	Number of trees	Size of the Random Forest
$\mathbf{w}$, b	Weight vector and bias	Parameters of the SVM hyperplane
ξ_i	Slack variable	Margin violation of observation i
$\phi(\cdot)$, $K(\cdot,\cdot)$	Feature map and kernel function	Linear or RBF kernel of the SVM
γ	Kernel coefficient	Width parameter of the RBF kernel
w_k	Class weight	$n/(Kn_k)$ for class k
μ_j, σ_j	Mean and standard deviation of feature j	Used for standardization

INTRODUCTION

Problem Statement

Direct marketing campaigns are costly when they are sent to customers who are unlikely to respond. In insurance, a predictive model that ranks customers by their likelihood of holding a particular product allows the company to contact only the most promising customers and to use its marketing budget more effectively. The CoIL Challenge 2000 was designed around this problem: it asked participants to predict which customers hold a caravan insurance policy and to describe the characteristics of the policyholders [1]. This study addresses the prediction task only.

The study uses the Insurance Company Benchmark (COIL 2000) dataset, which describes 5,822 labelled customers by 85 predictors and a binary target, `CARAVAN`, that indicates caravan policy ownership. The predictors combine socio-demographic information about the customer's postcode area with the customer's premiums and policy counts for insurance products. A separate set of 4,000 customers is provided for evaluation, and its labels were released after the competition [1].

The main difficulty is the imbalance of the target. Only 348 of the 5,822 labelled customers (about 6%) own a caravan policy. Under this distribution, accuracy is not a useful measure: a classifier that predicts non-ownership for every customer reaches about 94% accuracy without identifying a single policyholder. The original challenge therefore evaluated models by the number of policyholders found among a ranked selection, and this study follows that ranking perspective. Average Precision is used as the primary metric for model selection, with ROC-AUC as a complementary measure of ranking quality.

The data also present other statistical challenges. Previous analysis of this benchmark found that the variance of the learning algorithms, rather than their bias, dominates the prediction error, and that feature selection improves the performance of most algorithms [2]. These findings suggest that flexible models using all predictors may overfit the small number of policyholders, and they motivate the regularized models, the feature-selection experiments and the evaluation protocol used here.

Accordingly, this study compares five machine learning models for predicting caravan policy ownership and examines how the representation of the predictors affects their performance. The official training data are divided into development and validation subsets, models are compared on the validation set, and the official test set

is used only once, for the final model. The results are interpreted as predictions of policy ownership, not as the effect of a marketing mailing on a customer's response.

Structure of the report

The remainder of this report is organized into four chapters and a conclusion.

Chapter 1 defines the business problem and formulates it as a supervised learning task. It describes the structure of the COIL 2000 dataset and its predictor groups, discusses the class imbalance and the other statistical challenges, and reviews the literature on the CoIL Challenge 2000 that motivates the evaluation protocol.

Chapter 2 presents the exploratory data analysis and the feature preparation. After the stratified split into development and validation subsets, it checks data quality, tests the association between each predictor and policy ownership with chi-square tests and Benjamini–Hochberg correction, examines the shape of these associations, and removes eight redundant columns. Five non-monotonic predictors with the highest Information Value, together with the nominal variable `MOSTYPE`, are then one-hot encoded, giving a 152-column representation.

Chapter 3 introduces the five classification algorithms: Logistic Regression, K-Nearest Neighbours, Decision Tree, Random Forest and Support Vector Machine. For each algorithm it explains the theoretical basis, the hyperparameters to be tuned and the required scaling. It also describes the experimental procedure: the grid search over 76 configurations, the evaluation metrics, the feature-selection experiments and the final model selection.

Chapter 4 reports and interprets the results. It compares the tuned models on the validation set, examines the effect of feature selection, and reports the performance of the selected model on the official test set. It closes with a discussion of the limitations of the study.

The Conclusion summarizes the main findings and suggests directions for future work.

CHAPTER 1: PROBLEM AND DATASET OVERVIEW

This chapter introduces the caravan insurance prediction problem based on the CoIL Challenge 2000 dataset. It describes the business context, formulates the task as a supervised learning problem, presents the structure of the data, and reviews the literature that defines both the benchmark objectives and the statistical difficulties of the task. The chapter also states the evaluation protocol adopted in this project.

1.1 Problem Definition

1.1.1 Business Problem

The CoIL Challenge 2000 originates from a classical direct-marketing problem: identifying customers who are likely to hold a caravan insurance policy. In direct-mail marketing, untargeted campaigns are expensive because most of the mailings reach customers who will not respond. A predictive model allows the company to rank its customers and contact only those with the highest scores, which improves the response rate and makes better use of the marketing budget.

The original benchmark consisted of two tasks:

1. **Prediction task:** identify the customers who are most likely to own a caravan policy.
2. **Description task:** profile the existing policyholders and derive interpretable rules that are useful to marketing specialists.

For the prediction task, participants received an unlabeled evaluation set of 4,000 customers and had to select the 800 customers (20%) expected to contain as many policy owners as possible. The score was the number of actual policy owners among the selected customers [1].

This project addresses the prediction task. The description task is not treated.

1.1.2 Machine Learning Formulation

Let the labelled dataset be

$$\mathscr{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N},$$

where $\mathbf{x}_i \in \mathbb{R}^{85}$ is the vector of the 85 input attributes of customer i, and y_i is the binary target

$$y_i = \begin{cases} 1, & \text{if customer } i \text{ holds a caravan policy,} \\ 0, & \text{otherwise.} \end{cases}$$

A learning algorithm estimates a scoring function $f(\mathbf{x}_i) = s_i \in \mathbb{R}$, which is either an estimate of the probability of ownership or a relative score. Customers are sorted in descending order of s_i, and the highest-ranked customers are selected for the campaign.

The target records *ownership* of a policy, not the *response* of a customer to a mailing. The original study uses ownership as a practical proxy for the behaviour of interest, while acknowledging that the two are not identical [2]. The task is therefore interpreted as predicting policy ownership, not as observing future purchasing behaviour after a marketing stimulus.

1.1.3 Prediction Objective and Evaluation Metrics

Accuracy is not an appropriate measure for this task because the positive class is rare. Approximately 6% of the customers own a caravan policy [2], so a classifier that predicts "non-owner" for every customer reaches about 94% accuracy without identifying a single owner.

The original challenge counted the owners found among the 800 selected customers. In this project the models are compared with metrics that evaluate the quality of the whole ranking:

- **Average Precision (AP):** the primary metric, used for model selection. It summarizes the precision–recall curve and is sensitive to how well the few positive cases are ranked near the top.
- **ROC-AUC:** a secondary, threshold-independent measure of the ability to rank owners above non-owners.
- **F1-score:** a threshold-dependent measure computed with each classifier's default decision rule and reported for reference only.

1.2 Dataset Overview

1.2.1 Dataset Source and Structure

The data were supplied by Sentient Machine Research for the CoIL Challenge 2000 and are available from the UCI KDD archive. They combine socio-demographic information derived from postal-code areas with information about customers' ownership of insurance products [1]. The official split consists of a labelled training file, an unlabelled evaluation file, and a separate file containing the evaluation labels, which were released after the competition.

Bảng 1.1 Structure of the CoIL Challenge 2000 dataset.

Property	Description
Training file	`ticdata2000.txt`
Training records	5,822
Attributes in the training file	86 (85 inputs + 1 target)
Evaluation (test) file	`ticeval2000.txt`
Evaluation records	4,000
Evaluation labels	`tictgts2000.txt`
Target attribute	`CARAVAN`

1.2.2 Feature Description

The 85 input attributes fall into three groups (Table 1.2). All of them are stored as integer codes, and their meaning must be read from the dataset documentation rather than from the numerical values.

Bảng 1.2 Groups of input attributes.

Group	Unit	No.	Coding
`M` (socio-demographic)	Postcode area	43	Mostly percentage bands of unequal width (0 = 0%, 1 = 1–10%, …, 9 = 100%). Exceptions include `MGEMLEEF` (age band) and `MKOOPKLA` (purchasing-power class). `MOSTYPE` (41 customer subtypes) and `MOSHOOFD` (10 main types) are nominal.
`P` (premium)	Customer	21	Contribution to a product category in bands: 0 = none, 1 = f1–49, …, 9 = at least f20,000.
`A` (policy count)	Customer	21	Number of policies held in the corresponding product category.

The socio-demographic variables describe the *neighbourhood* of a customer and are shared by all customers of the same postcode area. They are not direct measurements of the individual customer, and conclusions drawn from them should take into account the risk of ecological inference. The `P` and `A` variables describe the customer himself or herself.

1.2.3 Target Variable and Class Distribution

The target `CARAVAN` equals 1 for a policy owner and 0 otherwise. In the 5,822 training records, 348 customers (about 6%) are owners; the evaluation set has a similar prevalence of 5.95% (238 owners among 4,000 customers). With this imbalance, random selection of 800 customers is expected to find only about 48 owners, and high accuracy can be obtained by ignoring the minority class. The evaluation therefore has to be based on ranking quality, as described in the problem definition above.

1.2.4 Dataset Characteristics and Challenges

The bias–variance analysis of Van der Putten and van Someren describes the CoIL problem as a realistic task with noisy, correlated, redundant and relatively high-dimensional inputs and a weak relationship between inputs and target [2]. The main challenges are:

- **Class imbalance:** owners are rare, so accuracy is misleading and the number of positive cases available for training and validation is small (about 278 in the development subset and about 70 in the validation subset used in this project).
- **Redundant and correlated predictors:** several attributes carry overlapping information. Chapter 2 shows that some columns are exact functions of others.
- **Weak signal and noise:** the association between customer attributes and ownership is limited, which makes model fitting and the comparison of models unstable.
- **Coded and mixed variable types:** nominal, ordinal and count variables are all stored as integers, so the numerical codes cannot always be treated as continuous measurements.
- **Ranking requirement:** the application requires a useful ordering of customers, not only a class label.

1.3 Literature Review

1.3.1 The CoIL Challenge 2000 Synthesis

Van der Putten and van Someren [1] documented the benchmark and compared 43 submissions based on a variety of techniques, including decision trees, neural networks, Bayesian methods and genetic algorithms. The best submission found 121 of the 238 policy owners in the evaluation set within its 800 selected customers. The comparison showed that results depended strongly on how the problem was formulated and how the attributes were handled, and not only on the complexity of the learning algorithm.

1.3.2 Bias–Variance Analysis and Feature Selection

In a follow-up study, Van der Putten and van Someren [2] analysed the errors of learning algorithms on this data. They found that the variance component of the error, rather than the bias, dominates in a noisy problem of this kind. They also reported that feature selection improved the performance of most of the algorithms examined, which suggests that restricting model complexity and the number of predictors can be more useful than using flexible models with all attributes.

1.3.3 Implications for This Project

These studies motivate the methodology adopted in the following chapters:

1. Models are evaluated with ranking-oriented metrics (Average Precision and ROC-AUC) instead of accuracy.
2. Redundant predictors are identified and removed, and the effect of reducing the number of predictors is tested explicitly through feature-selection experiments.
3. Regularised and low-complexity models are included in the comparison, and class imbalance is handled through class weights.
4. Results are interpreted as predicting policy ownership, not as the effect of a marketing mailing.
5. The official test set is used only once, after all modelling decisions have been made, because the weak signal makes repeated use of the test set especially prone to optimistic conclusions.

CHAPTER 2: EXPLORATORY DATA ANALYSIS

This chapter presents the exploratory data analysis (EDA) and the feature preparation carried out on the CoIL Challenge 2000 data. Its objectives are to assess data quality, to study the association between the predictors and caravan insurance ownership, to identify redundant variables, and to derive the feature representation used for modelling in Chapter 3.

Because the target is highly imbalanced and the predictors are coded integers of different types, the analysis emphasizes statistically controlled feature screening, the shape of the predictor–target relationships, and the removal of redundant information. All supervised analyses and all data-driven preprocessing decisions are made on the *development subset* only, so that the validation set and the official test set remain independent of these decisions.

2.1 Data Preprocessing and Quality Assessment

2.1.1 Dataset Partitioning and Structural Validation

The study keeps the official CoIL 2000 split: 5,822 labelled training records and 4,000 official test records [1]. Each record has 85 predictors and the binary target `CARAVAN`. Immediately after loading, and before any supervised analysis or learned preprocessing, the official training set is divided into a development subset (80%) and a validation subset (20%) by stratified random sampling with a fixed random seed (42).

Bång 2.1 Dataset partitioning and analytical roles.

Partition	Records	Predictors	Positive rate	Purpose
Development set	4,657	85	5.97%	EDA, preprocessing, training
Validation set	1,165	85	6.01%	Model selection and tuning
Official test set	4,000	85	5.95%	Final evaluation only

Stratification keeps the positive rate almost identical in the development and validation subsets. The test labels are loaded from the separate official label file but are not inspected or used until the final evaluation. Feature screening, redundancy removal, encoding and model selection use the development data (and, for model comparison, the validation data); the official training set is used as a whole only for the final refit after the model has been selected.

The 85 predictors form three groups, which are used in the subsequent analysis:

- **Socio-demographic variables (`M`, 43 variables)**, describing the postcode area of

the customer, mostly as percentage bands;

- **Premium variables (`P`, 21 variables)**, giving the premium band for each insurance product;

- **Policy-count variables (`A`, 21 variables)**, giving the number of policies held for each product.

The training and test files have an identical schema (same column names and order), so every transformation can be applied consistently to all partitions.

2.1.2 Data Completeness and Quality Assessment

No missing values are present in the development set or in the official test features (0 missing values in total), so no imputation is needed. All predictors are stored as integers, but these integers are codes for nominal categories, ordinal bands or counts rather than continuous measurements. Large values are therefore not treated as statistical outliers, and no outlier removal is performed; the original coded observations are kept and their relationship with the target is examined directly in the following sections.

2.1.3 Target Class Distribution

Bång 2.2 Target class distribution in the development set.

Class	Label	Records	Percentage
Non-policyholders	0	4,379	94.03%
Policyholders	1	278	5.97%
Total	–	4,657	100.00%

Only 278 of the 4,657 development records (5.97%) own a caravan policy. Accuracy is therefore uninformative, and the modelling framework uses Average Precision (AP) as the primary metric, ROC-AUC as a complementary ranking measure, and F1-score for reference. No oversampling or undersampling is applied; the imbalance is handled in the models through the `class_weight='balanced'` option, where the algorithm supports it (Chapter 3).

2.2 Exploratory Analysis of Predictor Relationships

2.2.1 Univariate Association with Caravan Ownership

For each of the 85 predictors, a chi-square test of independence is applied to the contingency table of the predictor levels and the target:

$$H_0 : X_j \perp \text{CARAVAN} \qquad \text{versus} \qquad H_1 : X_j \not\perp \text{CARAVAN}.$$

Testing 85 hypotheses at the level $\alpha = 0.05$ would produce roughly four false positives by chance alone. The p-values are therefore adjusted with the Benjamini–Hochberg (BH) procedure, which controls the false discovery rate (FDR), and a predictor is considered significant when its adjusted q-value satisfies

$$\boxed{q_j < 0.05}.$$

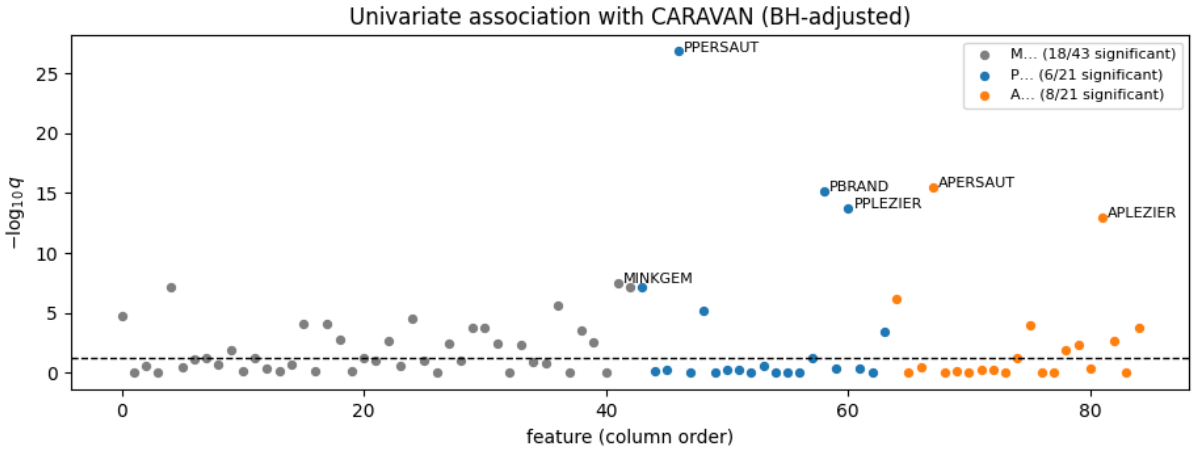


Figure 2.1 BH-adjusted association ($-\log_{10} q$) of each predictor with `CARAVAN`. The dashed line marks $q = 0.05$; the legend gives the number of significant predictors in each group.

Thirty-two predictors are significant after the correction: 18 of the 43 `M` variables, 6 of the 21 `P` variables and 8 of the 21 `A` variables (Figure 2.1). The strongest associations belong to product variables, in particular the car-insurance premium (`PPERSAUT`), the number of car policies (`APERSAUT`), the fire-insurance premium (`PBRAND`) and the boat-insurance premium (`PPLEZIER`). Several neighbourhood variables are also significant, including purchasing power (`MKOOPKLA`), the customer types (`MOSTYPE`, `MOSHOOFD`), income-related variables (`MINKM30`, `MINK4575`, `MINK7512` and `MINKGEM`). Most of the remaining product variables are not significant, which is probably related to the small number of customers who hold the corresponding products.

Statistical significance does not imply strong predictive power, and the chi-square approximation is less reliable for categories with very few observations. The q-values are therefore used for screening and ranking, not as a measure of effect size.

2.2.2 Shape of the Predictor–Target Associations

A significant association does not show whether ownership increases or decreases steadily with the level of the predictor. For the ordinal and count-coded predictors, the ownership rate $\widehat{P}(\text{CARAVAN} = 1 \mid X_j = x_{j,k})$ is computed for each level $x_{j,k}$ with at least five observations, and the Spearman correlation between the level and its ownership rate

is calculated:

$$r_j = \rho_{\text{Spearman}}\Big(x_{j,k},\ \widehat{P}(\text{CARAVAN} = 1 \mid X_j = x_{j,k})\Big)\,.$$

A predictor is described as clearly monotonic when $|r_j| \geq 0.8$. The nominal variables `MOSTYPE` and `MOSHOOFD` are excluded because their codes have no order. This is a descriptive heuristic, not a formal test of monotonicity.

Of the 32 significant predictors, 2 are nominal, 13 are classified as clearly monotonic, and **17 are not clearly monotonic**. Among the monotonic predictors are `MKOOPKLA` ($r = 0.91$), `MAUT1` ($r = 0.96$), `MHKOOP` ($r = 0.85$) and `MOPLLAAG` ($r = -0.88$). Figure 2.2 illustrates the different patterns. Ownership is about 11% for `PPERSAUT=6` but only about 2–3% for levels 0 and 5. For `PBRAND`, it peaks at level 4 (about 12%), is about 4% at level 0 and falls below 1% at level 6. For `MKOOPKLA`, it generally increases with purchasing power, peaks at level 7 (about 13%) and declines at the highest level 8 (about 9%). The last example also shows that the correlation threshold is a coarse summary: `MKOOPKLA` is classified as monotonic although its highest level departs from the trend.

The monotonicity heuristic is also less reliable for predictors with very few levels. For `AWAOREG`, `APLEZIER`, `AFIETS` and `ABYSTAND`, the correlation equals ± 1 because only a few levels pass the five-observation filter, so their "monotonic" label carries little information. `PWAPART` ($r = 0.80$) lies exactly at the boundary, and the computed value falls just below the threshold, so it is classified as not clearly monotonic.

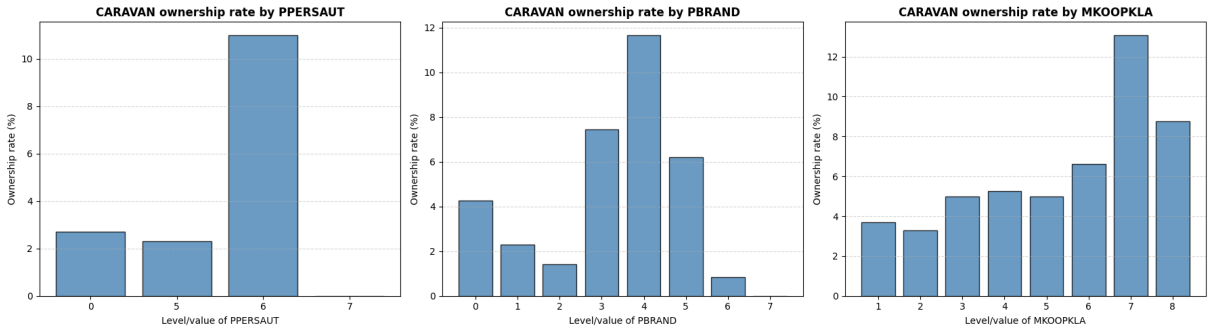


Figure 2.2 Observed caravan ownership rate by predictor level in the development set (levels with at least five observations).

2.2.3 Identifying Redundant Predictors

Several variables describe closely related products or hierarchical categories, so some columns may contain exactly the same information as others. Using the development set, a predictor X_a is said to *determine* another predictor X_b when every observed value of X_a corresponds to exactly one value of X_b:

$$\big|\{X_b \mid X_a = x\}\big| = 1 \quad \text{for every observed value } x \text{ of } X_a.$$

All ordered pairs of predictors are checked, and 10 directed perfect-dependency relationships are found (Table 2.3). For example, MOSTYPE determines MOSHOOFD, because each customer subtype belongs to exactly one main type, and the premium band of several products determines the corresponding policy count.

Bång 2.3 Perfect-dependency pairs found in the development set and the column removed.

Determining variable	Determined variable	Column removed
MOSTYPE	MOSHOOFD	MOSHOOFD
PWALAND	AWALAND	AWALAND
PVRAAUT	AVRAAUT	AVRAAUT
AVRAAUT	PVRAAUT	– (pair skipped)
PPERSONG	APERSONG	APERSONG
PGEZONG	AGEZONG	AGEZONG
PZEILPL	AZEILPL	AZEILPL
AZEILPL	PZEILPL	– (pair skipped)
PINBOED	AINBOED	AINBOED
AFIETS	PFIETS	PFIETS

From each directed pair, the determined (dependent) column is removed, unless one of the two columns has already been removed. This yields 8 distinct columns: AGEZONG, AINBOED, APERSONG, AVRAAUT, AWALAND, AZEILPL, MOSHOOFD and PFIETS. The pairs (PVRAAUT, AVRAAUT) and (PZEILPL, AZEILPL) determine each other. The rule removes only one member of each such pair, so the policy-count variables AVRAAUT and AZEILPL are removed, while the premium variables PVRAAUT and PZEILPL are kept. The delivery-van and surfboard categories therefore remain represented through their premiums. Neither AVRAAUT nor AZEILPL is among the 32 significant predictors, so the removal is not expected to affect the significance screening.

The same 8 columns are removed from the development, validation, full training and official test matrices. The removal decision is based on the development data only, and the resulting development matrix has **77 predictors**.

2.3 Feature Engineering and Representation

2.3.1 Treatment of Non-Monotonic Predictors

Many of the significant predictors do not follow a monotonic pattern. A model that treats such a predictor as a numerical variable, such as logistic regression or an SVM with a linear kernel, implicitly assumes that moving from one code to the next changes the log-odds by a constant amount. This assumption is inappropriate for PPERSAUT

or PBRAND, where the ownership rate rises and falls across levels. One-hot encoding represents each level separately and removes this assumption.

Encoding all 17 non-monotonic predictors would, however, add many columns (most of them have 7–10 levels) and increase the risk of overfitting on a small positive class. Information Value (IV) is therefore used to select the predictors that deserve the more flexible representation.

2.3.2 Information Value-Based Assessment

For each level i of a predictor, the Weight of Evidence (WOE) is

$$\mathrm{WOE}_i = \ln\left(\frac{P(X = i \mid Y = 1)}{P(X = i \mid Y = 0)}\right),$$

and the Information Value of the predictor is

$$\mathrm{IV} = \sum_i \left[P(X = i \mid Y = 1) - P(X = i \mid Y = 0)\right] \mathrm{WOE}_i.$$

A small smoothing constant (10^{-4}) is added to the counts so that the logarithm is defined when a level contains no observation of one class. IV is computed on the development subset only, after the 8 redundant columns have been removed. The conventional interpretation is given in Table 2.4.

Bång 2.4 Conventional interpretation of Information Value.

Information Value	Predictive interpretation
$\mathrm{IV} < 0.02$	Little or no predictive power
$0.02 \leq \mathrm{IV} < 0.10$	Weak
$0.10 \leq \mathrm{IV} < 0.30$	Medium
$0.30 \leq \mathrm{IV} < 0.50$	Strong
$\mathrm{IV} \geq 0.50$	Suspiciously high

These ranges are guidelines, not decision rules. Over all 77 retained predictors, the two highest IV values are those of PPERSAUT (0.619) and MOSTYPE (0.600), both above 0.5. For PPERSAUT this reflects the strong association shown in Figure 2.2. For MOSTYPE the value is probably inflated in part because the variable has 41 nominal levels, and IV generally increases with the number of levels even when the signal is weak. In both cases the high IV is a reason for caution, not evidence of leakage, since these variables are known before the target and the IV is computed only on development data.

2.3.3 Selection of Predictors for One-Hot Encoding

All 17 non-monotonic predictors remain in the 77-column matrix after redundancy removal. Table 2.5 gives their number of levels, Spearman correlation and IV, ranked by

IV.

Bång 2.5 Non-monotonic candidate predictors ranked by Information Value (development set).

Rank	Predictor	Levels	Spearman r	IV	Strength
1	PPERSAUT	5	−0.40	0.619	Suspiciously high
2	APERSAUT	7	−0.56	0.411	Strong
3	PBRAND	9	−0.33	0.408	Strong
4	MINKGEM	10	0.45	0.307	Strong
5	MSKA	10	0.30	0.206	Medium
6	PWAPART	4	0.80	0.144	Medium
7	MINK4575	10	0.25	0.137	Medium
8	AWAPART	3	−0.50	0.131	Medium
9	MOPLHOOG	10	0.65	0.124	Medium
10	PPLEZIER	7	0.50	0.114	Medium
11	MBERARBG	10	0.25	0.113	Medium
12	PMOTSCO	6	−0.20	0.106	Medium
13	MINK7512	10	0.18	0.104	Medium
14	ABRAND	7	−0.40	0.099	Weak
15	MSKC	10	−0.36	0.095	Weak
16	MBERHOOG	10	0.71	0.091	Weak
17	PBYSTAND	4	0.20	0.044	Weak

The five predictors with the highest IV are selected for one-hot encoding:

- PPERSAUT: car insurance premium;
- APERSAUT: number of car insurance policies;
- PBRAND: fire insurance premium;
- MINKGEM: average income level of the neighbourhood;
- MSKA: neighbourhood indicator for the social class A share.

The encoder is fitted on the development subset only and then applied unchanged to the validation and official test features; levels that were not seen during fitting are ignored (`handle_unknown='ignore'`). The five variables expand into $5+7+9+10+10=41$ indicator columns.

The nominal variable MOSTYPE is also one-hot encoded, regardless of its IV rank. Its codes have no order, so an integer representation would impose one on the 41 customer subtypes, which would be inappropriate for the linear models and the SVM. In the development set 40 subtypes are observed, which adds 40 indicator columns. Subtypes not seen during fitting are ignored at transformation time.

In total, the six encoded variables are replaced by 81 indicator columns, so the feature matrix grows from 77 to $77-6+81=152$ columns. All other predictors keep their original integer coding, and MOSHOOFD, which is determined by MOSTYPE, has already been removed.

2.4 Summary of Exploratory Findings and Preprocessing Decisions

Bång 2.6 Summary of EDA findings and the resulting preprocessing decisions.

Aspect	Empirical finding	Decision
Missing values	None in the development set or the official test features.	No imputation.
Class distribution	Positive rate of 5.97% in the development set.	Use AP as the primary metric; use balanced class weights in the models.
Statistical association	32 of 85 predictors have BH-adjusted $q < 0.05$.	Use q-values for feature screening (Chapter 3 experiments).
Monotonicity	17 significant predictors are not clearly monotonic.	Use one-hot encoding for the five with the highest IV; keep the others as integer codes.
Redundancy	10 directed perfect-dependency pairs.	Remove 8 columns (85 $\rightarrow$ 77 predictors).
Representation	Five non-monotonic predictors with the highest IV, and the nominal MOSTYPE.	One-hot encode these six variables (77 $\rightarrow$ 152 columns).
Leakage control	All supervised decisions use development data only.	Fit encoders and scalers on development data and apply them without refitting.

The exploratory analysis reduces the original 85 predictors to 77 through redundancy screening, and a selective one-hot encoding then expands the representation to 152 columns. The resulting feature matrix retains the information of the original attributes, removes exactly duplicated information, and avoids imposing a numerical order on the predictors whose relationship with the target is clearly non-monotonic or nominal. These decisions form the common input of all models in Chapter 3.

CHAPTER 3: MACHINE LEARNING MODELS FOR PREDICTION

This chapter presents the theoretical foundations and the implementation procedure of the five classification algorithms used to predict caravan insurance ownership. Each algorithm represents a different learning mechanism and has hyperparameters that control its complexity, regularization, neighbourhood structure or decision boundary.

The hyperparameters of every model are searched on a grid using the validation Average Precision (AP) as the selection criterion. Additional experiments examine whether restricting the predictors according to statistical significance and Information Value (IV) improves the models. The final model is chosen from the development and validation results only, refitted on the complete official training set, and evaluated once on the official test set. The results of these steps are reported and discussed in Chapter 4.

3.1 List of Models

Five supervised classifiers are investigated: Logistic Regression, K-Nearest Neighbours (KNN), Decision Tree, Random Forest and Support Vector Machine (SVM). They cover linear statistical classification, distance-based learning, rule-based partitioning, ensemble learning and maximum-margin classification. Logistic Regression is an interpretable linear baseline, KNN captures local patterns among similar customers, the Decision Tree represents nonlinear relationships through sequential splits, the Random Forest averages many such trees, and the SVM seeks a boundary with maximum margin.

The target `CARAVAN` is binary (1 = the customer owns caravan insurance). Since the positive class is rare, the models are compared mainly by AP, with ROC-AUC as a secondary metric and F1-score as a threshold-dependent reference.

3.2 Fundamental Theory of the Models

3.2.1 Logistic Regression

Logistic Regression estimates the probability of the positive class from a linear combination of the predictors $\mathbf{x} = (x_1, \ldots, x_p)$:

$$z = \beta_0 + \sum_{j=1}^{p} \beta_j x_j, \qquad P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + \exp(-z)}.$$

The parameters are estimated by minimizing the binary cross-entropy loss

$$\mathscr{L}(\beta) = -\frac{1}{n} \sum_{i=1}^{n} \Big[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \Big],$$

plus a penalty on the size of the coefficients:

$$\mathscr{L}_{L1} = \mathscr{L}(\beta) + \lambda \sum_{j=1}^{p} |\beta_j|, \qquad \mathscr{L}_{L2} = \mathscr{L}(\beta) + \lambda \sum_{j=1}^{p} \beta_j^2.$$

The L1 (Lasso) penalty can set some coefficients exactly to zero and therefore performs implicit feature selection. The L2 (Ridge) penalty shrinks the coefficients towards zero without eliminating them and stabilizes the estimates when predictors are correlated.

In the implementation the inverse regularization strength $C \propto 1/\lambda$ is used, so a smaller C means stronger regularization. The searched hyperparameters are:

- **Regularization strength C:** $0.01, 0.1, 1, 10$.
- **Penalty:** L1 and L2.

The `liblinear` solver is used in the search because it supports both penalties; it is fixed, not tuned. The option `class_weight='balanced'` gives more weight to the minority class.

Because the penalty acts on coefficient magnitudes, the predictors are standardized with the `StandardScaler`:

$$x_j^* = \frac{x_j - \mu_j}{\sigma_j},$$

where μ_j and σ_j are estimated on the development training set.

3.2.2 K-Nearest Neighbours (KNN)

KNN is a non-parametric, instance-based method. To classify a new observation $\mathbf{x}$, it finds the k closest training observations $\mathscr{N}_k(\mathbf{x})$ and predicts by a vote of their labels. With uniform weights, the predicted class is $\hat{y} = \text{mode}\{y_i : i \in \mathscr{N}_k(\mathbf{x})\}$, and the estimated probability of ownership is the fraction of owners among the neighbours. With distance weights, closer neighbours receive larger weights. Two distances are considered:

$$d_E(\mathbf{x}, \mathbf{z}) = \sqrt{\sum_{j=1}^{p} (x_j - z_j)^2}, \qquad d_M(\mathbf{x}, \mathbf{z}) = \sum_{j=1}^{p} |x_j - z_j|.$$

The searched hyperparameters are:

- **Number of neighbours k:** $3, 5, 7, 9, 11, 13, 15$. A small k gives a flexible boundary that is sensitive to noise; a large k gives smoother predictions but may hide local structure.
- **Weighting:** `uniform` or `distance`.

- **Distance metric:** Euclidean or Manhattan.

Since predictions depend directly on distances, the features are scaled to $[0,1]$ with the `MinMaxScaler`,

$$x_j^* = \frac{x_j - x_{j,\min}}{x_{j,\max} - x_{j,\min}},$$

using the minimum and maximum of the development training set.

3.2.3 Decision Tree

A Decision Tree recursively splits the feature space. Each internal node tests a predictor, each branch is an outcome of the test, and each leaf gives a class probability. The split at a node is chosen to maximize the impurity reduction

$$\Delta I = I(t) - \left[\frac{N_L}{N_t} I(t_L) + \frac{N_R}{N_t} I(t_R)\right],$$

where $I(\cdot)$ is the node impurity and N_t, N_L, N_R are the numbers of observations in the parent and the two child nodes. Two impurity measures are compared:

$$\text{Gini}(t) = 1 - \sum_{k=1}^{K} p_k^2, \qquad H(t) = -\sum_{k=1}^{K} p_k \log_2 p_k.$$

The searched hyperparameters are:

- **Splitting criterion:** `gini` or `entropy`.
- **Maximum depth:** $3, 5, 8, 12, 15, 20, 25$ and `None` (no explicit limit). Shallow trees are simple but may miss structure; deep trees can memorize noise.

Splits depend only on the ordering of values within a feature, so scaling is not needed and the tree is trained on the one-hot encoded but unscaled data.

3.2.4 Random Forest

A Random Forest averages B decision trees, each trained on a bootstrap sample and each considering only a random subset of predictors at every split:

$$\hat{P}(Y = 1 \mid \mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} \hat{P}_b(Y = 1 \mid \mathbf{x}).$$

The randomness makes the trees diverse, and averaging reduces the variance of a single tree. The searched hyperparameters are:

- **Number of trees:** 100 and 200.

- **Maximum depth:** $5, 10, 15$ and `None`.

- **Maximum features per split:** `sqrt` ($\approx \sqrt{p}$) and `log2` ($\approx \log_2 p$). Considering fewer predictors increases the diversity of the trees but may prevent a tree from finding useful splits.

As for a single tree, the forest uses the one-hot encoded but unscaled data.

3.2.5 Support Vector Machine (SVM)

A linear SVM looks for the hyperplane $\mathbf{w}^T\mathbf{x} + b = 0$ that maximizes the margin to the nearest observations of each class (the support vectors). For data that cannot be separated perfectly, the soft-margin problem is

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \quad \text{s.t.} \quad y_i\big(\mathbf{w}^T\phi(\mathbf{x}_i) + b\big) \geq 1 - \xi_i, \;\; \xi_i \geq 0,$$

where ξ_i is the margin violation of observation i and C controls its penalty: a smaller C gives a wider margin and stronger regularization. Nonlinear boundaries are obtained with kernels:

$$K_{\text{linear}}(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T\mathbf{x}_j, \qquad K_{\text{RBF}}(\mathbf{x}_i, \mathbf{x}_j) = \exp\big(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2\big).$$

The searched hyperparameters are:

- **Regularization parameter C:** $0.01, 0.1, 1, 10$.

- **Kernel:** `linear` and `rbf`.

The RBF parameter γ is not tuned and keeps the default value of the implementation (`gamma='scale'`). The SVM is trained on standardized features, and `probability=True` is set so that probability estimates are available for the AP and ROC-AUC calculations.

3.3 Detail of the Applying Process

3.3.1 Data Partitioning and Experimental Design

The 5,822 labelled training records are split by stratified sampling (80/20, seed 42) into a development set of 4,657 records and a validation set of 1,165 records, as described in Chapter 2. Models and preprocessing transformations are fitted on the development set, hyperparameters and model families are compared on the validation set, and the 4,000-record official test set is reserved for the final evaluation. Each tuning configuration is therefore evaluated on a single development–validation split, not by cross-validation.

3.3.2 *Feature Representation and Scaling*

All models start from the 152-column matrix described in Chapter 2: eight redundant columns are removed, and the five non-monotonic predictors with the highest IV together with the nominal `MOSTYPE` are one-hot encoded. The encoder and the scalers are fitted on the development set only and then applied to the validation and test features without refitting. The scaling depends on the model (Table 3.1).

Bång 3.1 Input data and fixed settings of each model.

Model	Scaling	Class weight	Other fixed settings
Logistic Regression	StandardScaler	balanced	`liblinear` solver in the search, `max_iter`=5000
KNN	MinMaxScaler	not available	–
Decision Tree	none	balanced	random seed 42
Random Forest	none	balanced	random seed 42
SVM	StandardScaler	balanced	`probability=True`, default γ

3.3.3 *Hyperparameter Tuning Strategy*

For each model family, the candidate values are combined into a full Cartesian grid by explicit iteration. Each combination is trained on the development set and evaluated on the validation set.

In total, $8 + 28 + 16 + 16 + 8 = 76$ configurations are evaluated. For the imbalanced target, `class_weight='balanced'` is used for Logistic Regression, Decision Tree, Random Forest and SVM; it assigns class k the weight

$$w_k = \frac{n}{K\, n_k},$$

where n is the number of training observations, $K = 2$ the number of classes and n_k the number in class k, so that errors on the minority class are penalized more. KNN has no such option in this implementation; its predictions depend only on the labels (and optional distance weights) of the neighbours.

3.3.4 *Evaluation Metrics and Model Selection*

The primary metric is Average Precision,

$$\mathrm{AP} = \sum_n (R_n - R_{n-1}) P_n, \qquad \mathrm{Precision} = \frac{TP}{TP+FP}, \quad \mathrm{Recall} = \frac{TP}{TP+FN},$$

where P_n and R_n are the precision and recall at the n-th threshold. AP evaluates how well the few positive customers are ranked, which suits the rare-event setting. ROC-AUC, the

Bång 3.2 Hyperparameter search spaces.

Model	Hyperparameter	Candidate values	Configs
Logistic Regression	C	0.01, 0.1, 1, 10	8
	Penalty	L1, L2	
KNN	Neighbours k	3, 5, 7, 9, 11, 13, 15	28
	Weighting	Uniform, Distance	
	Metric	Euclidean, Manhattan	
Decision Tree	Criterion	Gini, Entropy	16
	Maximum depth	3, 5, 8, 12, 15, 20, 25, None	
Random Forest	Number of trees	100, 200	16
	Maximum depth	5, 10, 15, None	
	Maximum features	sqrt, log2	
SVM	C	0.01, 0.1, 1, 10	8
	Kernel	Linear, RBF	

area under the ROC curve, is the secondary metric and measures the probability that a random owner is scored above a random non-owner. The F1-score,

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}},$$

depends on a classification threshold; it is computed with each classifier's default decision rule and is not optimized, so it is reported only for reference. Training AP is also recorded to monitor overfitting.

For each model family, the configuration with the highest validation AP is its best configuration, and the families are then compared through their best validation results.

3.3.5 Feature Engineering Experiments

Three further experiments examine whether using fewer predictors helps. The chi-square tests with Benjamini–Hochberg adjustment (Chapter 2) identify 32 significant predictors; one of them (`MOSHOOFD`) has been removed as redundant, so 31 significant predictors remain. Their IV values, computed on the development set, are split at the 33.33rd and 66.67th percentiles (0.1061 and 0.1443) into a high-IV group of 10 predictors, a medium-IV group of 11 and a low-IV group of 10. The experiments are:

- **E1 (High only):** the 10 high-IV predictors, which include `MOSTYPE`;
- **E2 (High + Medium):** the 21 predictors of the first two groups;

- **E3 (High + Medium + Low):** all 31 significant predictors.

In every experiment, the five models are retrained with the best hyperparameters found in the full-feature search. The one-hot encoding is applied to whichever of the six encoded variables (`PPERSAUT`, `APERSAUT`, `PBRAND`, `MINKGEM`, `MSKA` and `MOSTYPE`) belong to the subset, while the other predictors keep their integer codes. The encoder and scalers are refitted on the development data of the subset and applied to the validation data. Their results are compared with the full-feature configuration of the previous step on the same validation set.

3.3.6 Final Model Selection and Test-Set Evaluation

The model family with the highest best validation AP in the full-feature hyperparameter search is selected, together with its best configuration. The preprocessing pipeline (encoder and, where required, scaler) is refitted on all 5,822 official training records, the selected model is retrained on the same data, and the fitted pipeline transforms the official test features. Test performance is measured by AP and ROC-AUC. The test set is used only in this last step: it does not influence the choice of the model family, the hyperparameters or the preprocessing.

CHAPTER 4: MODEL ANALYSIS

This chapter reports and interprets the results of the experiments described in Chapter 3. All comparisons between models and configurations use the validation set, and the official test set is used once for the final model. Because only about 70 of the 1,165 validation records are positive, differences of one or two hundredths in AP between configurations are within the range of sampling noise and should be read as indications, not as firm rankings.

4.1 Baseline Model Performance Analysis

Five algorithms were tuned on the same development and validation data: Logistic Regression, Random Forest, SVM, Decision Tree and KNN. Average Precision (AP) is the primary metric and ROC-AUC the complementary ranking metric. Here, "baseline" refers to the best configuration of each family found with the full 152-column feature set, before any feature selection.

Bång 4.1 Best configuration of each model family (full feature set), ranked by validation AP.

Model	Train AP	Val. AP	Val. ROC-AUC	Val. F1
Logistic Regression	0.2160	0.2769	0.8245	0.2368
Random Forest	0.3081	0.2440	0.8052	0.2543
SVM	0.2188	0.2158	0.7865	0.2133
Decision Tree	0.1568	0.1700	0.7701	0.2163
KNN	0.2279	0.1613	0.7140	0.0000

Bång 4.2 Selected hyperparameters of each family.

Model	Best configuration
Logistic Regression	$C = 0.01$, L1 penalty (`liblinear`)
Random Forest	100 trees, maximum depth 5, `max_features=sqrt`
SVM	$C = 0.01$, RBF kernel
Decision Tree	entropy criterion, maximum depth 3
KNN	$k = 15$, uniform weights, Manhattan distance

Logistic Regression has the highest validation AP (0.2769) and ROC-AUC (0.8245), followed by Random Forest (0.2440) and SVM (0.2158). Decision Tree (0.1700) and KNN (0.1613) are clearly weaker, and KNN has the lowest AP and the lowest ROC-

AUC (0.7140). Its F1-score is zero because, with 15 neighbours and uniform weights, it never predicts the positive class at the default decision rule.

The tuning results show a consistent pattern: *the simplest or most strongly regularized configuration of each family performs best.*

- Logistic Regression is best with the strongest penalty ($C = 0.01$, L1). The second-best setting, $C = 0.01$ with L2, reaches 0.2038, and the remaining settings lie between 0.1986 and 0.2016.

- The Decision Tree is best at depth 3. The unrestricted tree has a training AP of 0.988 but a validation AP of only 0.070 with the Gini criterion (0.064 with entropy).

- The Random Forest is best at depth 5 with 100 trees, whereas forests with unlimited depth reach a training AP of about 0.93 and a validation AP of only 0.124–0.126.

- The SVM with an RBF kernel is best with the smallest C. With $C = 10$ the training AP rises to 0.863 and the validation AP falls to 0.098. The linear kernel is weaker at every value of C, with a best validation AP of 0.193.

- KNN improves with the number of neighbours, and its best $k = 15$ is the largest value in the grid, so the optimum may lie beyond the searched range. KNN with distance weighting reaches a training AP of about 0.99, because every training point is its own nearest neighbour, and it does not generalize.

These results agree with the literature discussed in Chapter 1: in a noisy problem with a weak signal and few positive cases, model flexibility mainly increases variance.

The difference between training and validation AP gives further information. KNN (+0.067) and Random Forest (+0.064) have training scores clearly above their validation scores, which indicates overfitting. SVM (+0.003) and Decision Tree (-0.013) show almost no gap, and Logistic Regression has a validation AP higher than its training AP (-0.061). A validation score above the training score is not a sign of a better model. With about 70 positives, AP on the validation set is noisy, and the value of 0.2769 is the maximum over eight configurations, so it is probably optimistic. The jump from 0.2038 (the second-best Logistic Regression configuration) to 0.2769 should be interpreted with this in mind. A Logistic Regression with the default setting ($C = 1$, L2 penalty, `lbfgs` solver, unscaled features) has a validation AP of only 0.2002, which shows how much of the final result comes from regularization and scaling.

4.2 Feature Selection Experiment Analysis

After the baseline search, three experiments restricted the predictors to the statistically significant ones. After the Benjamini–Hochberg correction and the removal of the redundant column `MOSHOOFD`, 31 significant predictors remained. They were divided into three groups by IV: 10 high-IV, 11 medium-IV and 10 low-IV predictors. The experiments use the high group (E1), the high and medium groups (E2), and all 31 predictors (E3), with the hyperparameters of Table 4.2. The one-hot encoding is applied to those of the six encoded variables that belong to each subset.

Bång 4.3 Validation Average Precision under the feature-selection experiments. The best value of each model is in bold.

Model	Baseline (152 cols.)	E1	E2	E3
Logistic Regression	0.2769	0.2603	0.2716	**0.2804**
Random Forest	**0.2440**	0.2276	0.2297	0.2385
SVM	0.2158	0.2220	**0.2342**	0.2257
Decision Tree	0.1700	**0.1785**	0.1694	0.1692
KNN	**0.1613**	0.1383	0.1475	0.1533

Bång 4.4 Validation ROC-AUC under the feature-selection experiments. The best value of each model is in bold.

Model	Baseline (152 cols.)	E1	E2	E3
Logistic Regression	0.8245	0.8135	0.8266	**0.8278**
Random Forest	0.8052	0.8097	**0.8109**	0.8050
SVM	0.7865	0.7955	**0.7976**	0.7800
Decision Tree	0.7701	**0.7826**	0.7647	0.7592
KNN	**0.7140**	0.6752	0.6825	0.6981

The effect of feature selection depends on the algorithm.

- **Logistic Regression** is best with all 31 significant predictors (E3: AP 0.2804, ROC-AUC 0.8278). The gain over the baseline is only 0.0035 in AP and 0.0033 in ROC-AUC, so feature selection leaves its performance essentially unchanged while using fewer predictors.

- **SVM** is best with the 21 predictors of E2 (AP 0.2342). All three subsets exceed the baseline AP of 0.2158, but the largest subset (E3, 0.2257) is below E2, so the medium-IV predictors add information while the low-IV predictors do not.

- **Decision Tree** reaches its highest AP under E1 (0.1785 against 0.1700 for the baseline), and E1 also gives the highest ROC-AUC among its experiments (0.7826). A plausible explanation, consistent with the variance argument of Chapter 1, is that the high-IV subset removes noisy predictors on which the tree would otherwise split. The present experiments do not test this directly.

- **Random Forest** and **KNN** do not improve in any experiment. The full feature set gives the best AP for both, and the Random Forest's best ROC-AUC is obtained under E2 (0.8109), which is only 0.006 above its baseline.

Using more predictors is thus not uniformly better or worse. Two cautions apply: the hyperparameters of all experiments were tuned on the full feature set, and the same validation set was used for all comparisons, so the experiments are not independent confirmations. The differences between experiments for a single model are mostly of the order of 0.01, which is smaller than the plausible sampling error.

4.3 Final Model Evaluation

The model family was selected on the validation AP of the baseline search, and the winner is Logistic Regression ($C = 0.01$, L1). The encoder, scaler and model were refitted on all 5,822 official training records, and the model was evaluated once on the 4,000 official test records, which were not used in any selection step.

Bång 4.5 Final Logistic Regression performance.

Metric	Validation	Official test
Average Precision	0.2769	0.1532
ROC-AUC	0.8245	0.7278

On the official test set the model reaches AP = 0.1532 and ROC-AUC = 0.7278. Both are lower than on the validation set. The test AP is about 2.6 times the positive rate of the test set (5.95%, the AP of a random ranking), so the model ranks owners above non-owners better than chance, but the predictive power is modest.

Several factors can contribute to the gap between validation and test results, and the available evidence does not allow them to be separated. The validation AP is the maximum over several configurations and is estimated from only about 70 positives, so it is optimistically biased and noisy. The test AP is itself subject to sampling variation, and the final model is refitted on a larger training set than the one on which the hyperparameters were chosen. A difference between the data distributions of the two samples is also possible. The size of the gap alone does not establish data leakage, and the proce-

dure was designed so that none of the preprocessing or model-selection decisions used test information.

4.4 Summary of Model Analysis

The analysis leads to the following findings.

1. Logistic Regression with strong L1 regularization had the best validation AP (0.2769) and ROC-AUC (0.8245), followed by Random Forest (0.2440) and SVM (0.2158); Decision Tree and KNN were the weakest.
2. In every family the best configurations are the simple, strongly regularized ones, and flexible configurations overfit the small positive class.
3. The effect of feature selection depends on the model: Logistic Regression is best with all 31 significant predictors, SVM with the 21 predictors of E2, and the Decision Tree with the 10 high-IV predictors. Random Forest and KNN do best with the full feature set. The differences are generally small.
4. The final Logistic Regression achieved AP = 0.1532 and ROC-AUC = 0.7278 on the official test set, above the 5.95% prevalence but well below the validation values.

The study has several limitations that should be taken into account when interpreting these results.

- Models were compared on a single development–validation split with few positive cases; repeated stratified cross-validation would give more reliable estimates and a measure of uncertainty.
- The experiments of Section 4.2 reuse hyperparameters tuned on the full feature set and the same validation set.
- The nominal variable `MOSTYPE` (41 subtypes, 40 of them observed in the development set) was one-hot encoded, which adds 40 sparse columns to a model fitted on about 280 positive cases. Alternatives such as target or frequency encoding inside the training folds were not tested.
- The redundancy rule removed the policy-count variables `AVRAAUT` and `AZEILPL`, keeping the premium variables `PVRAAUT` and `PZEILPL`. The delivery-van and surfboard categories are therefore represented only through their premiums.
- The F1-score was computed at default thresholds, and the original CoIL objective (the number of owners among the 800 highest-ranked of 4,000 customers) was not evaluated explicitly.

CONCLUSION

This project studied the prediction of caravan insurance ownership on the CoIL Challenge 2000 data, a problem characterized by a rare positive class (about 6%), coded predictors of mixed types, and a weak and noisy relationship between predictors and target. The official training set was divided into a development and a validation subset before any supervised analysis, and the official test set was used only once, for the final evaluation.

Main findings. The exploratory analysis found no missing values and 32 of the 85 predictors significantly associated with ownership after Benjamini–Hochberg correction, with car and fire insurance variables, the car-policy count and neighbourhood purchasing power among the strongest. Seventeen of the significant predictors were not clearly monotonic. Eight columns were exact functions of other columns and were removed, leaving 77 predictors. The five non-monotonic predictors with the highest Information Value, together with the nominal customer type `MOSTYPE`, were one-hot encoded, giving a 152-column representation.

Five classifiers were tuned with Average Precision as the selection criterion. Logistic Regression with strong L1 regularization was the best model (validation AP 0.2769, ROC-AUC 0.8245), ahead of Random Forest (AP 0.2440) and SVM (AP 0.2158), while Decision Tree and KNN performed worst. In every family the simple and strongly regularized configurations generalized best and flexible configurations overfit, which agrees with the variance-dominated behaviour reported for this benchmark. Experiments with significant predictors grouped by Information Value showed that the best number of predictors depends on the algorithm: Logistic Regression was best with all 31 significant predictors, SVM with the 21 highest-IV and medium-IV predictors, and the Decision Tree with the 10 highest-IV predictors. Random Forest and KNN did best with the full feature set. All differences were small.

The final Logistic Regression, refitted on all 5,822 training records, obtained AP = 0.1532 and ROC-AUC = 0.7278 on the 4,000 official test records. This is about 2.6 times the test prevalence (5.95%) and shows that the model ranks potential policyholders better than chance, but the performance is lower than on the validation set and modest in absolute terms. The gap can be explained in part by the optimism and noise of a validation estimate based on about 70 positive cases and a selection among several configurations; it does not by itself indicate leakage, since all preprocessing and selection decisions were based on development and validation data only.

Limitations. The comparison relied on one stratified development–validation split

rather than repeated cross-validation, so the uncertainty of the reported scores is not quantified. The feature-selection experiments reused hyperparameters tuned on the full feature set and the same validation set. The nominal variable `MOSTYPE` was one-hot encoded, which adds many sparse columns to a small positive class. The redundancy rule removed the policy-count variables of two mutually dependent product pairs and kept their premium variables, so these products are represented only through their premiums. The target measures ownership rather than response to a mailing, and the socio-demographic variables describe postcode areas rather than individuals, so the results should not be read as the effect of a marketing action on a particular customer.

Future work. Possible extensions are repeated stratified cross-validation with confidence intervals for the model comparison; alternative encodings of `MOSTYPE`, such as target encoding inside the training folds; a wider search for KNN (larger k, since the best value lies at the edge of the grid) and for the RBF parameter γ of the SVM; gradient-boosted trees and calibrated probabilities; threshold selection tied to a campaign budget; and evaluation with the original CoIL criterion, the number of policy owners among the 800 highest-ranked customers of the 4,000 evaluation records, to compare the results directly with the published benchmark.

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